

JEE MAINS GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C)

Time: 3HRS

Date:19-01-24

Max. Marks: 300

Name of the Student: _____

H.T. NO:

19-01-24_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-21(N)_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

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PHYSICS

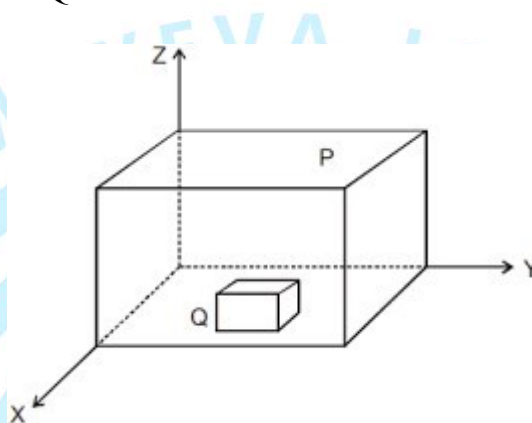
MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

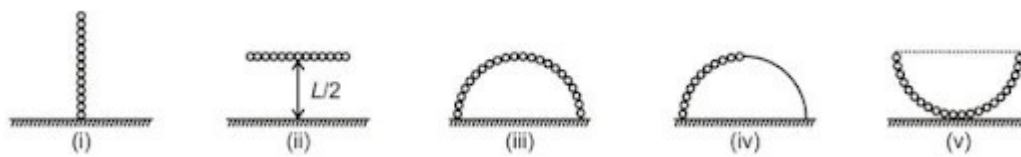
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A hollow cubical box P is moving on a smooth horizontal surface in x-y plane with constant acceleration of $\vec{a} = 3\hat{i} + 4\hat{j} \text{ m/s}^2$. A block Q of mass 2Kg is at rest inside the cubical box as shown in fig. Coefficient of friction between P and Q is $\mu = 0.6$. Then the friction force between P&Q is

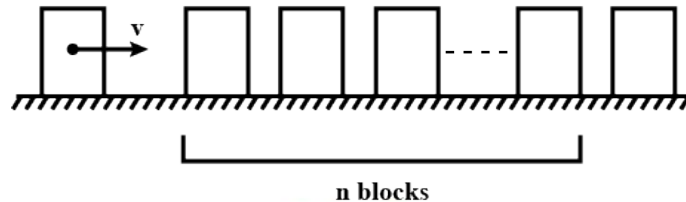


- A) 5N B) 8N C) 12N D) 10N
2. A chain of length L and mass M is arranged as shown in following five cases. In fig iv chain covers half of the semi-circle. The correct decreasing order of potential energy is. (Assumed zero at horizontal surface)

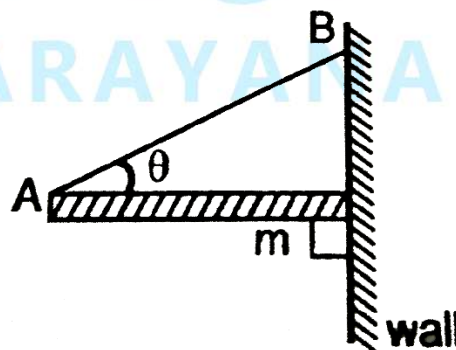


- A) I > ii > iii > IV > V B) I = ii > iii > IV > V
- C) I = ii > IV > iii > V D) I = ii > iii > V > IV

3. A set of 'n' identical cubical blocks lies at rest parallel to each other along a line on a smooth horizontal surface. The separation between the near surfaces of any two adjacent blocks is L. The block of one end is given by a speed V towards the next one at time $t=0$. All collisions are completely inelastic, then the last block starts moving at

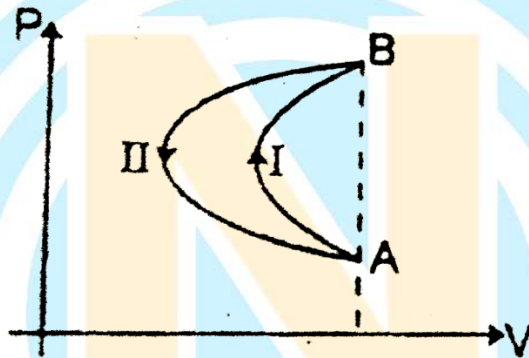


- A) $t = (n-1)L/v$ B) $t = (n-2)(n-1)L/2v$
 C) $t = (n-1)L/2v$ D) $t = n(n-1)L/2v$
4. Moment of Inertia of a uniform quarter disc of radius R and mass M about an axis through its Centre of mass and perpendicular to its plane is:
- A) $\frac{MR^2}{2} - M \left(\frac{4R}{3\pi} \right)^2$ B) $\frac{MR^2}{2} - M \left(\frac{4\sqrt{2}R}{3\pi} \right)^2$
 C) $\frac{MR^2}{2} + M \left(\frac{4R}{3\pi} \right)^2$ D) $\frac{MR^2}{2} + M \left(\frac{4\sqrt{2}R}{3\pi} \right)^2$
5. A rod of mass m is supported by string AB and friction due to wall. Then friction force on rod due to wall is : (g = acceleration due to gravity).



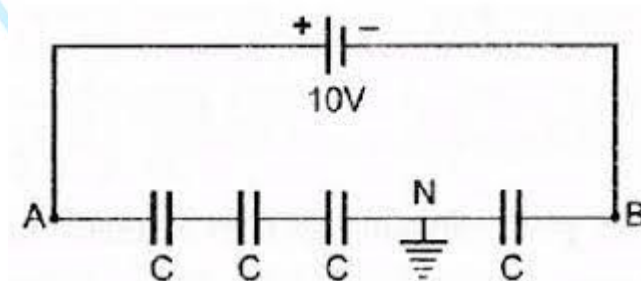
- A) mg upward B) mg downward C) $\frac{mg}{2}$ upward D) $\frac{mg}{2}$ downward

6. Radius of a planet is n times the radius of earth R . A satellite revolves around it in a circle of radius ' $4nR$ ' with angular velocity ' ω '. Then the acceleration due to gravity on the planet's surface is
- A) $R\omega^2$ B) $16R\omega^2$ C) $32nR\omega^2$ D) $64nR\omega^2$
7. A uniform rod of length L has a mass per unit length λ and area of cross-section A . The elongation in the rod is l , due to its own weight if it is suspended from the ceiling of a room vertically. The Young's modulus of the rod is
- A) $\frac{2\lambda g L^2}{Al}$ B) $\frac{\lambda g L^2}{2Al}$ C) $\frac{2\lambda g L}{Al}$ D) $\frac{\lambda g L^2}{Al}$
8. In a cyclic process, a gas is taken from state A to state B via the path –I as shown in indicator diagram and taken back to state A via the path –ii. In the complete cycle.



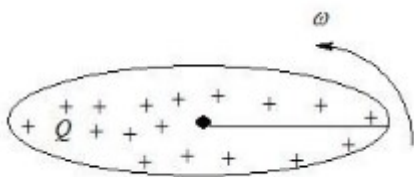
- A) work is done on the gas B) heat is given to the gas
- C) No work done by the gas D) nothing can be said about work done
9. Assertion (A): if a point charge be rotated in a circle around a charge, the work done will be zero.
- Reason (R): Work done is equal to dot product of force and displacement.
- A) Both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- B) Both Assertion and Reason are correct but the Reason is not correct explanation of the Assertion.
- C) Assertion is correct Reason is incorrect.
- D) If Both Assertion and reason are incorrect.

10. A particle of mass 10 gm, is placed in potential field given by $V = (50x^2 + 100) \text{ J/Kg}$. Then the frequency of oscillation in cycle/sec is
- A) $5/\pi$ B) $10/\pi$ C) $1/\pi$ D) $\pi/5$
11. At $t=0$ a transvers wave pulse travelling in the positive x direction with a speed of 2m/s in a long wire is described by the function $y = \frac{6}{x^2}$, given that $x \neq 0$. Transverse velocity of particle at $x=2\text{m}$ and $t=2\text{s}$ may be:
- A) 7m/s B) -3m/s C) 8m/s D) -1.5m/s
12. Regarding meter bridge experiment:
- Statement 1: For proper connection of circuit, the deflection of galvanometer must be opposite when jockey touches at the end points of wire
- Statement 2: Jockey is dragged along the wire, instead of point to point contact to get null point.
- A) Statement 1 is true and statement 2 is false
- B) Statement 1 is false and statement 2 is true
- C) Both the statements are true
- D) Both the statements are false
13. Four identical capacitors are connected in series with a 10V cell as shown in the fig. The point N is grounded. The potential of point A is

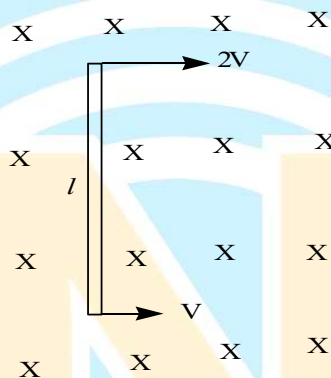


- A) 10V B) 7.5V C) 2.5V D) -2.5V

14. A thin disc of mass m , radius R and charge Q uniformly distributed over it. Which is rotates with an angular speed ω . Then the magnetic induction at the Centre of sheet is



- A) $\frac{\mu_0 Q \omega}{4\pi R}$ B) $\frac{\mu_0 Q \omega}{2\pi R}$ C) $\frac{2\mu_0 Q \omega}{\pi R}$ D) $\frac{\mu_0 Q \omega}{\pi R}$
15. In the figure magnetic field points in to the plain of paper and the rod of length l is moving in this field such that the bottom most point has a velocity v and the top most point has the velocity $2v$. The emf induced across the end points of the rod is given by



- A) Bvl B) $2Bvl$ C) $\frac{3}{2}Bvl$ D) $\frac{1}{2}Bvl$
16. An astronaut floating freely in space decides to use his flash light as a rocket. He shines a 10 watt light beam in a fixed direction so that he acquires a momentum in the opposite direction. If his mass is 80 Kg, how long must he need to reach a velocity of 1m/s.
- A) 9 sec B) 2.4×10^3 sec C) 2.4×10^6 sec D) 2.4×10^9 sec
17. Match the following

Light beam	Nature of object/image formed
i) Incident diverging beam	P) Real Image
ii) reflected/refracted converging beam	Q) Real object
iii) Incident converging beam	R) Virtual image
iv) Reflected diverging beam	S) Virtual object

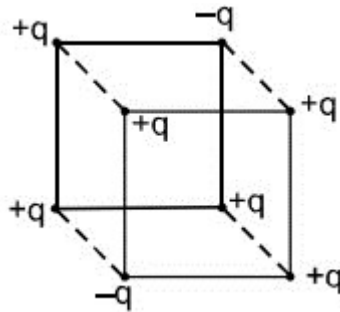
A) i-Q; ii-P; iii-S; iv-R

B) i-P; ii-Q; iii-S; iv-R

C) i-S; ii-R; iii-Q; iv-P

D) i-R; ii-S; iii-P; iv-Q

18. The side of the cube is $\sqrt{3}\text{cm}$ and point charges are kept at each corner as shown in diagram. Interaction electrostatic potential energy of all the charges is (in joules) when $q = 10\mu\text{C}$.



- A) 30J B) 100J C) 120J D) zero
19. A sensor is exposed for time 't' to a lamp of power P placed at a distance l. The sensor has a small opening that is 4d in diameter. Assuming all energy of the lamp is given off as light, the number of photons entering the sensor if the wavelength of light is λ . (c speed of light in vacuum).
- A) $N = \frac{P\lambda d^2 t}{hcl^2}$ B) $N = \frac{4P\lambda d^2 t}{hCl^2}$
- C) $N = \frac{P\lambda d^2 t}{4hcl^2}$ D) $N = \frac{P\lambda d^2 t}{16hcl^2}$
20. Assertion: The drift current in a p-n junction is from the n side to the P side
Reason: drift current opposite in direction to the diffusion current
- A) If both Assertion & Reason are correct and the Reason is correct explanation of the Assertion
- B) If both Assertion & Reason are correct but the Reason is not correct explanation of the Assertion
- C) If the Assertion is correct but Reason is incorrect
- D) Both Assertion and Reason are incorrect

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

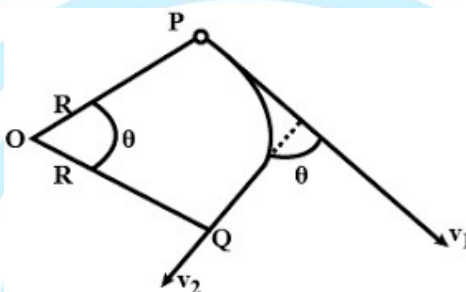
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Marking scheme: +4 for correct answer, -1 in all other cases.

21. The relative density of a ball can be found by hanging the ball from a spring balance and noting the reading. The balance reads $4.00 \pm 0.02\text{N}$ in air and $1.00 \pm 0.02\text{N}$ in water. The maximum percentage error in measurement of relative density is $X/6$. Where X is

22. A particle describes an angle θ in a circular path with a constant speed v . Then the average acceleration of the particle during the motion in the arc is given by $\frac{NV^2}{R\theta} \sin\left(\frac{\theta}{2}\right)$.

Where N is

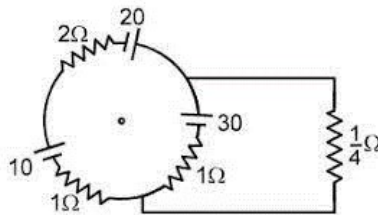


23. Two persons of masses 55 Kg and 65 Kg respectively are at the opposite ends of a boat. The length of the boat is 3 m and weighs 100 Kg. The 55 Kg man walks up to the 65 Kg man and sits with him. If the boat is in still water the Centre of mass of the system shifts by (in meters)

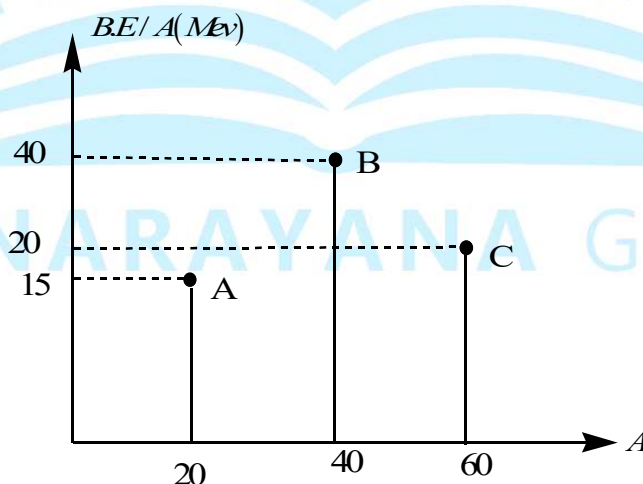
24. Water is flowing through a horizontal pipe of variable cross-section. The rate of volume flow is 0.2lit/sec . The pressure is 30 000 Pa at a point where the area of cross-section is $5 \times 10^{-5}\text{m}^2$. The pressure at a point where the area of cross-section is 10^{-4}m^2 is $N \times 1000$ Pa. where N is

25. The amount of heat supplied (in calories) to decrease the volume of an ice water mixture by 1cm^3 Without any change in temperature, is equal to (density of ice 0.9 gm/cc and latent of heat of ice is 80 cal/gm).

26. In the following circuit diagram, the current through the resistor $\frac{1}{4}\Omega$ is in ampere



27. A radio set operates at 4V DC. A transformer with 18 turns in the secondary coil is used to step down the input 220V AC emf to 6V AC emf. This AC emf is then rectified by another circuit to give 4V DC which is fed to the radio. Find the number of turns in the primary.
28. In the Young's double-slit experiment, when a glass-plate ($R.I = 1.5$) of thickness t is introduced in the path of one of the interfering beams (wavelength λ), the intensity at the position where the central maximum occurred previously remains unchanged. The minimum thickness of the glass- plate in multiple of λ is
29. If the kinetic energy of the particle is increased to 16 times its previous value, the percentage change (magnitude) in de-Broglie wavelength of the particle is
30. The graph shows $B.E/A$ values for three hypothetical elements A, B and C. The energy released when C disintegrates into A and B is (in Mev).

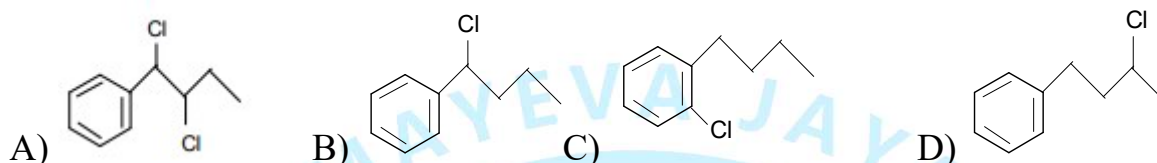


SECTION – I (SINGLE CORRECT ANSWER TYPE)

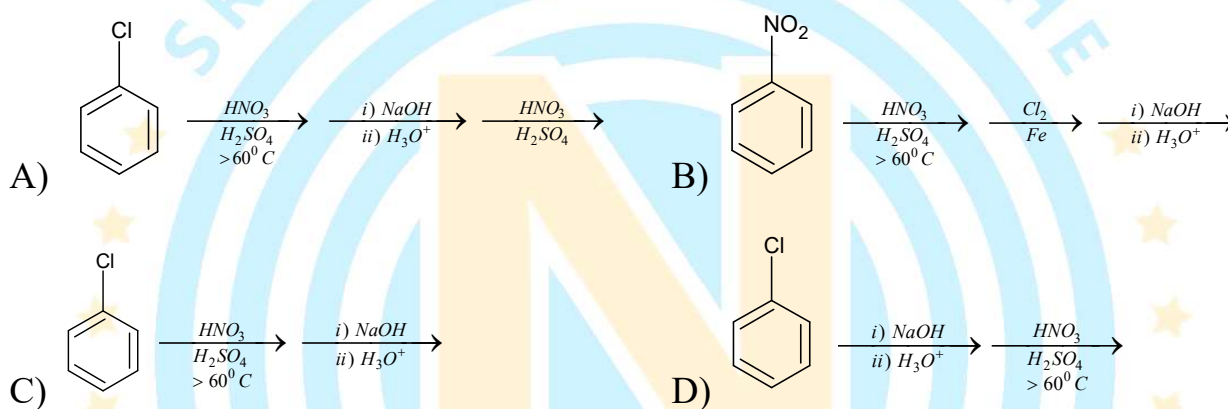
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31. $Ph-C \equiv C-CH_2-CH_3 \xrightarrow[H_2SO_4/Hg^{2+}]{H_2O} A \xrightarrow[HCl]{Zn/Hg} B \xrightarrow{Cl_2/h\nu} C$ The compound (C) (major product) is



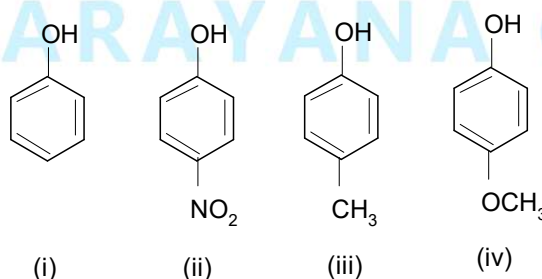
32. Which of the following methods will give rise to picric acid



33. The correct order of reactivities of the following alkyl halides for an SN2 reaction is

- A) $R-F > R-Cl > R-Br > R-I$ B) $R-F > R-Br > R-Cl > R-I$
C) $R-Cl > R-Br > R-F > R-I$ D) $R-I > R-Br > R-Cl > R-F$

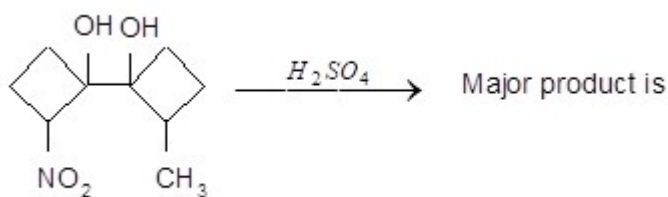
34. In the following compounds,



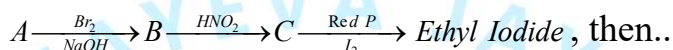
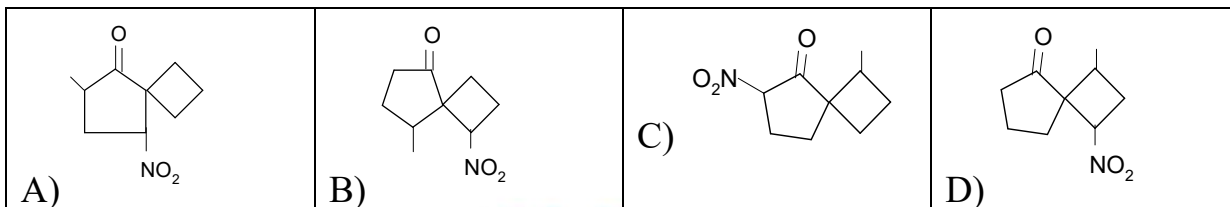
The correct order of acidity

- A) $ii > iv > i > iii$ B) $ii > i > iii > iv$ C) $iv > ii > i > iii$ D) $ii > iii > i > iv$

35.



36.

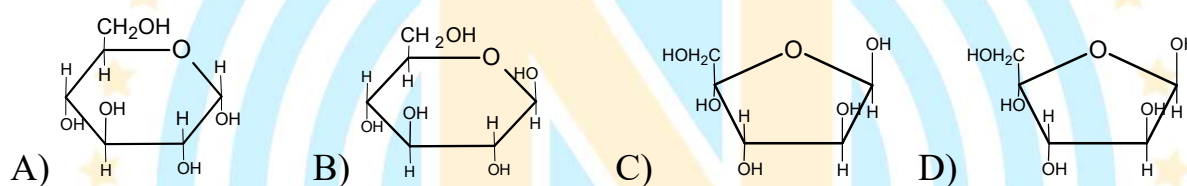


A) A is Ethyl amine

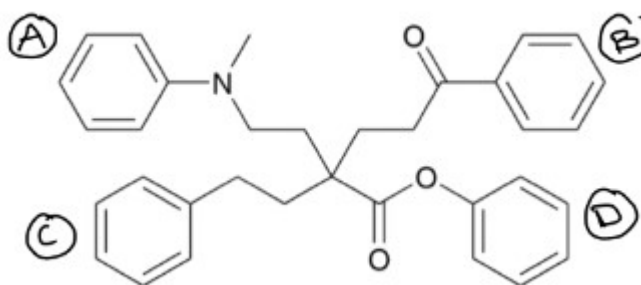
B) A is Ethanamide

C) A is Propanamide

D) A is Propyl halide

37. The structure of β - D glucopyranose is

38. The following compound has four aromatic rings. Rank them in terms of increasing reactivity towards EAS.



A) B < C < A < D

B) C < B < D < A

C) A < D < C < B

D) B < D < C < A

39. Which of the following reaction is not possible?

A) $2\text{KBr} + \text{Cl}_2 \rightarrow 2\text{KCl} + \text{Br}_2$ B) $2\text{NaCl} + \text{F}_2 \rightarrow 2\text{NaF} + \text{Cl}_2$ C) $2\text{KI} + \text{Br}_2 \rightarrow 2\text{KBr} + \text{I}_2$ D) $2\text{KCl} + \text{I}_2 \rightarrow 2\text{KI} + \text{Cl}_2$

40. The chromyl chloride test fails in case of chlorides of all the following group of metals
- A) Hg, Sn, Pb, Ag B) Cu, Fe, Ba, Mg
C) Hg, Co, Cd, Ag D) Fe, Co, Ni, Pb
41. The spin only magnetic moment of $[Mn(SCN)_6]^{-4}$ complex is nearly 6 B.M. Then it's configuration is
- A) $t_{2g}^4 e_g^1$ B) $t_{2g}^5 e_g^0$ C) $t_{2g}^3 e_g^2$ D) $t_{2g}^2 e_g^3$
42. Colourless – tetrahedral complexes among the following is
- A) $K_2[Cu(CN)_4]$ B) $Na_2[NiCl_4]$ C) $K[BF_4]$ D) $Ni(CO)_4$
43. No. of unpaired electrons in B_2 molecule is
- A) 2 B) 6 C) 8 D) 10
44. Choose the correct statement
- A) CCl_4 is non polar because C- Cl bond is non polar
B) CCl_4 is polar because C – Cl bond is polar
C) Though C – Cl bond is polar, CCl_4 is non polar
D) Though C – Cl bond is non-polar, CCl_4 is polar
45. The molecular shapes of SF_4 , CF_4 and XeF_4 are
- A) The same with 2, 0 and 1 lone pairs of electrons on the central atom respectively
B) The same with 1, 1 and 1 lone pairs of electrons on the central atoms respectively
C) Different with 0, 1 and 2 lone pairs of electrons on the central atom respectively
D) Different with 1, 0 and 2 lone pairs of electrons on the central atom respectively
46. The hydrogen electrodes dipped in two solutions of pH = 3 and pH = 6 connected through a salt bridge produce a maximum emf of
- A) 0.177 V B) 0.3 V C) 0.052 V D) 0.104 V

47. A buffer solution is prepared by mixing 10 ml of 1 M CH_3COOH and 20 ml of 0.5 M CH_3COONa and then diluted to 100 ml with distilled water. If the pK_a of CH_3COOH is 4.76, what is the pH of the buffer solution prepared?
- A) 3.76 B) 4.76 C) 5.8 D) 6.4
48. If 0.2 gram of an organic compound containing carbon, hydrogen and oxygen on combustion, gave 0.147 g carbondioxide and 0.12 g of water. What will be the content of oxygen in the substance?
- A) 73.29 % B) 68.45% C) 85.23% D) 89.50%
49. Which of the following statements/relationships is not correct in thermodynamic changes?
- A) $q = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)
- B) For a system at constant volume, heat involved merely changes to internal energy
- C) $w = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible compression of an ideal gas)
- D) $\Delta U = 0$ (isothermal reversible compression of an ideal gas)
50. A 0.004 M solution of Na_2SO_4 is isotonic with a 0.010 M solution of glucose at same temperature. The apparent degree of dissociation of Na_2SO_4 is
- A) 25% B) 50% C) 75% D) 85%

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

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51. 0.405 g of an organic compound was kjeldahlised and ammonia evolved was absorbed into 50 ml of semi normal solution of sulphuric acid. The residual acid was diluted with distilled water and volume was made up to 150 ml. 20 ml of this acid solution required 31ml of decinormal NaOH for complete neutralization. Calculate the percentage of nitrogen in the compound.

52. The treatment of propyne with excess of CH_3MgI releases 4.48ml of a gas at STP. The mass of propyne (in mg) added is
53. How many of the following statements are correct?
- i) The rate of nitration of C_6H_6 is greater than that of C_6D_6
 - ii) Acid catalysed hydration of alkenes is a regioselective reaction
 - iii) Electrolysis of aqueous solution of sodium fumarate produces ethyne.
 - iv) Acylium ion is more stable than alkyl carbocation.
 - v) Benzene decolourises Baeyer's reagent.
 - vi) Carbon – carbon bond length in benzene is equal to that of ethene.
 - vii) Bromination at allylic position by NBS is a free radical mechanism.
54. How many moles of diborane is required for 6 moles of an alkene in Hydroboration Oxidation method.
55. The total number of nodes in a $5d_{x^2-y^2}$ orbital is...
56. The number of reducing sugars from the following are?
D-glucose, D-fructose, Maltose, Lactose, Sucrose, Starch, Cellulose.
57. Standard Enthalpy of neutralization of H_3PO_3 is -106.68 kJ/mol using NaOH. If standard enthalpy of neutralization of HCl by NaOH is -55.84 kJ/mol, calculate standard enthalpy of ionization of H_3PO_3 (in kJ) into its ions.
58. How many of the following are Acidic oxides
 $MnO, MnO_2, Mn_2O_7, Cr_2O_3, CrO_3, V_2O_5, CrO, V_2O_4, V_2O_3$
59. At $25^\circ C$, the solubility product of $Mg(OH)_2$ is 1.0×10^{-11} . At which P^{OH} will Mg^{2+} ions start precipitating in the form of $Mg(OH)_2$ from a solution of $0.001M$ Mg^{2+} ions?
60. At infinite dilution the molar conductance of Al^{+3} and SO_4^{-2} ion are 189 and $160 \Omega^{-1} cm^2 mol^{-1}$ respectively. Calculate the equivalent conductivity at infinite dilute of $Al_2(SO_4)_3$. (in C.G.S system)

MATHEMATICS

MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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61. If Z is a complex number satisfying $|\operatorname{Re}(z)| + |\operatorname{Im}(z)| = 4$ then $z = x + iy$ satisfies.

- A) $2 \leq |z| \leq 3$ B) $2\sqrt{2} \leq |z| \leq 3 - \sqrt{3}$ C) $2\sqrt{2} \leq |z| \leq 4$ D) $|z| \geq 4$

62. **Statement – 1 :** The system of equations $x + y + z = 6, x + 2y + 3z = 14, x + 4y + 7z = 30$ has infinitely many solutions

Because

Statement – 2: For a system of n -equations in n -unknowns if determinant value of the coefficient matrix be zero, then the system of equations has infinitely many solutions.

- A) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for Statement-1.
B) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for Statement-1.
C) Statement-1 is true, Statement-2 is false.
D) Statement-1 is false, Statement-2 is true.

63. Let $\alpha, \beta (\alpha > \beta)$ be the roots of the quadric equation $x^2 - x - 7 = 0$. If

$P_n = \alpha^n - \beta^n, n \in N$ then $\frac{P_{20}P_{21} - P_{19}P_{21} - P_{20}^2 + P_{19}P_{20}}{P_{18}P_{19}}$ equals

- A) 16 B) 25 C) 14 D) 49

64. The sum of all 3- digit numbers that are divisible by either 3 or 4 but not by 48 is

- A) 2,15,486 B) 2,05,486 C) 1,95,386 D) 2,37,114

65. Let $A = \{1, 2, 3, 4, 5, 6\}$. Then the probability that a randomly chosen function f from A to A satisfying $f(3) = 2f(1)$ is
- A) $\frac{1}{15}$ B) $\frac{1}{5}$ C) $\frac{1}{30}$ D) $\frac{1}{10}$
66. Let A, B be two bags. Bag A contains 3 Red and 4 black balls. Bag B contains 2 Red and 5 black balls. One of the bags A, B is chosen at random and a ball is drawn at random from it. If it is Red then the probability that the ball is chosen from bag A is
- A) $\frac{2}{5}$ B) $\frac{4}{5}$ C) $\frac{3}{5}$ D) $\frac{1}{5}$
67. Two fair dice are rolled once. Then the mean of sum of the numbers on them is
- A) 7 B) 7.5 C) 6.5 D) 8
68. The maximum value of the function $f(x) = 3x^3 - 18x^2 + 27x - 40$ on the set $\{x \in \mathbb{R} / x^2 + 30 \leq 11x\}$ is
- A) 122 B) -222 C) -122 D) 222
69. If $g(x) = x^2 + x - 1$ and $g \circ f(x) = 4x^2 - 10x + 5$, then $f\left(\frac{5}{4}\right)$ is equal to
- A) $-\frac{3}{2}$ B) $\frac{1}{2}$ C) $-\frac{1}{2}$ D) $\frac{3}{2}$
70. The range of $\sin^{-1} x + \cos^{-1} x + \tan^{-1} x$ is
- A) $(0, \pi)$ B) $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$ C) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ D) $(-\pi, \pi)$
71. A line having direction ratios 1: -4 : 2 intersect the lines $\frac{x-7}{3} = \frac{y-1}{-1} = \frac{z+2}{1}$ and $\frac{x}{2} = \frac{y-7}{3} = \frac{z}{1}$ at the points P and Q then $(PQ)^2$ equals.
- A) 42 B) 84 C) 24 D) 48

72. **Statement-1** : If three parallel chords of a circle have lengths 2, 3, 4 and subtend angle $x, y, x + y$ at the centre (where $x + y < 180^\circ$), then $\cos x = \frac{17}{32}$

Statement-2 : In a circle of radius r , if chord of length ℓ subtends an angle θ at the centre, then $\cos \theta = 1 - \frac{\ell^2}{2r^2}$. Also $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta \forall \alpha, \beta \in \mathbb{R}$

- A) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for Statement-1.
- B) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for Statement-1.
- C) Statement-1 is true, Statement-2 is false.
- D) Statement-1 is false, Statement-2 is true.

73. Let
$$\begin{vmatrix} 1+x & x & x^2 \\ x & 1+x & x^2 \\ x^2 & x & 1+x \end{vmatrix} = ax^5 + bx^4 + cx^3 + dx^2 + ex + f$$

COLUMN – I

COLUMN – II

- | | |
|------------------------|-------|
| a) Value of f is | p) 0 |
| b) Value of e is | q) 1 |
| c) Value of $a + c$ is | r) -1 |
| d) Value of $b + d$ is | s) 3 |
- A) a-q, b-s, c-r, d-q B) a-s, b-r, c-s, d-q
- C) a-q, b-r, c-s, d-r D) a-s, b-q, c-r, d-s

74. The sum of eccentricity and the length of transverse axis of the hyperbola

$$\frac{x^2}{16} - \frac{y^2}{9} = 1 \text{ is}$$

- A) $\frac{37}{4}$ B) 8 C) $\frac{29}{3}$ D) 25

75. If $f(x) = \int \frac{5x^8 + 7x^6}{(x^2 + 1 + 2x^7)^2} dx$ ($x \geq 0$) and $f(0) = 0$ then $f(1)$ equals

- A) $\frac{1}{2}$ B) $\frac{-1}{2}$ C) $\frac{1}{4}$ D) $\frac{-1}{4}$

76. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying

$$\int_0^{\pi/2} f(\sin 2x) \sin x \, dx = \lambda \int_0^{\pi/4} f(\cos 2x) \cos x \, dx$$
 Then the value of ' λ ' is

- A) 1 B) -1 C) $\sqrt{2}$ D) $-\sqrt{2}$

77. If $\frac{dy}{dx} = \frac{2^{x+y} - 2^x}{2^y}$, $y(0) = 1$ then $y(1)$ is

- A) $\log_2(1 + e^2)$ B) $\log_2(2e)$ C) $\log_2(2 + e)$ D) $\log_2(1 + e)$

78. If the coefficient of x^{50} in the expansion $\left(x^4 + \frac{3}{x}\right)^{15}$ is K then the number of distinct prime numbers that divide K is

- A) 2 B) 3 C) 4 D) 5

79. The value of $36(4\cos^2 9^\circ - 1)(4\cos^2 27^\circ - 1)(4\cos^2 81^\circ - 1)(4\cos^2 243^\circ - 1)$ is

- A) 36 B) 27 C) 18 D) 54

80. Let $f(x) = x \left[\frac{x}{2} \right]$ for $-10 < x < 10$, where $[t]$ denotes the greatest integer function.

Then the number of points of discontinuities of f is

- A) 4 B) 6 C) 8 D) 10

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. Let α, β, γ be the roots of $x^3 - 2x^2 + 2x - 1 = 0$ and let $a_n = \alpha^n + \beta^n + \gamma^n$ for $n \in N$ then the value of a_{162} is

82. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{pmatrix}$. Then the sum of the principle diagonal elements of the matrix

$(A + I)^{11}$ is equal to

83. A letter is taken at random from the letters of the word "PROBABILITY", If P is the probability of selected letter is a vowel then the value of $11p$ is

84. A stone is projected vertically upward moves such that its distance feet in time t sec is given by $s = 112t - 16t^2$ then the maximum height reached by the stone in (feet) is

85. If $\vec{a}_1, \vec{a}_2, \vec{a}_3, \dots, \vec{a}_{20}$ are 20 unit vectors such that $\sum_{i=1}^{20} \vec{a}_i = \vec{0}$ then the value of

$\sum_{0 \leq i < j \leq 20} \vec{a}_i \cdot \vec{a}_j$ is

86. Let $a, b \in \mathbb{R}$. If the mirror image of the point $P(a, 6, 9)$ with respect to

$\frac{x-3}{7} = \frac{y-2}{5} = \frac{z-1}{-9}$ is $(20, b, -a-9)$, then $|a+b|$ is equal to

87. Let B and C be two points on the line $x+y=0$ and are symmetric about the origin.

If A lies on $y-2x=2$ such that ABC is an equilateral triangle of area Δ then $3\Delta^2$ equals.

88. If A be the area of the region $\{(x, y) \in \mathbb{R} \times \mathbb{R} / |2x - 1| \leq y \leq |x^2 - x|, 0 \leq x \leq 1\}$.

Then the value of $(6A + 11)^2$ is equal to

89. If (α, β) be the nearest point on the parabola $y^2 = 8x + 2y - 9$ from the line $y = x + 4$

Then the sum of first ten terms of the arithmetic progression $\alpha, \alpha + \beta, \alpha + 2\beta, \dots$ is equal to

90. If the variance of the terms in an increasing A.P $a_1, a_2, a_3, \dots, a_{11}$ is 90 then its common difference is

